

AI Content Creator

Personal Brand Engine

Not generic AI — your voice, your frameworks, your proof points.

Ioanna Renta

VP Engineering · AI Course · Berlin

The Problem

Generic AI Output

"Build a culture of psychological safety and focus on clear communication."

Could have been written by anyone.
No specifics. No evidence. No point of view.
Sounds like every other LinkedIn post.

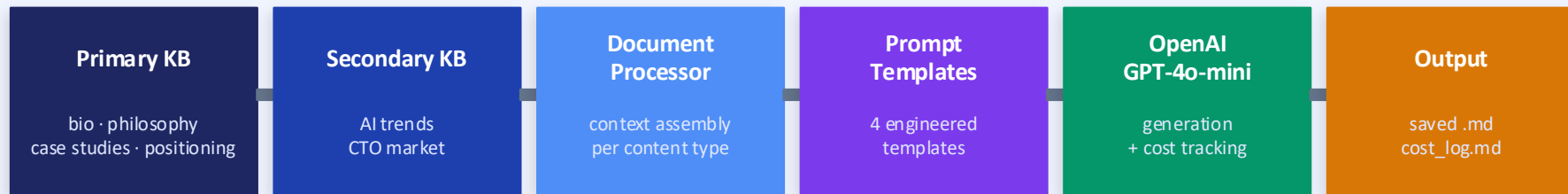
This System

"50% of the org voluntarily adopted QMI within six months. Voluntary adoption is the only honest signal — if people use it because they fear consequences, the data is corrupted."

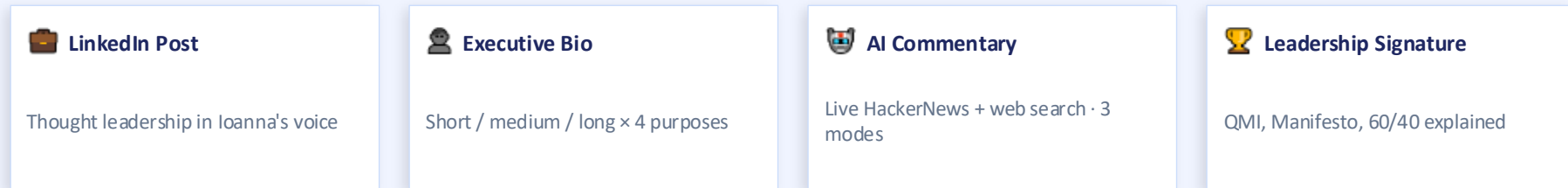
Specific. Evidence-based. Could only have been written by Ioanna.

The difference: specificity, evidence, and a point of view that only comes from real experience.

Solution Architecture



4 Content Types Generated:



trello.com/b/kteKFdzb

Web UI: Gradio · Live news: HackerNews Algolia API · All outputs auto-saved with cost log

Two Knowledge Bases

The knowledge base IS the product. The LLM is just the rendering engine.

PRIMARY KNOWLEDGE BASE

knowledge_base/primary/

bio.md

Career story · short / medium / long bios

leadership_philosophy.md

QMI · Engineering Manifesto · 60/40 model

case_studies.md

AUDI A8 · DH ML · sennder incidents

positioning.md

Target roles · voice guidelines · brand tone

SECONDARY RESEARCH LAYER

knowledge_base/secondary/

ai_trends.md

AI adoption in engineering orgs · governance gap

cto_market.md

VP/CTO compensation · hiring criteria · fractional market

Update policy: Primary files updated as career evolves · Secondary refreshed quarterly

Real Experience. Real Numbers.

The knowledge base is built from actual outcomes — not templates.

**20/week
→ 2/mo**

P1 incidents
reduced

55s → 5s

AUDI A8
cold start

50%

Org adopted
QMI voluntarily

+35%

Organic growth
via ML recs

100%

Customer
retention

0

Regrettable
attrition

sennder — Incident Reduction

P1 incidents: 20/week → 2/month
MTTR: 1hr → under 10 min
Blameless post-mortem culture + observability
infrastructure

Delivery Hero — QMI Framework

8-dimension quality maturity model
50% of 5,000-person org adopted voluntarily
35% organic growth + 60% retention via ML

HERE / AUDI A8 — Rescue

Customer relationship at risk, fixed delivery date
Shipped on production line date
Cold-start: 55s → 5s via tile caching

All data is real. All frameworks were implemented at scale.

Technical Implementation — 9 modules, 1 coherent pipeline

Pipeline Core

document_processor.py

Reads all KB markdown files. Assembles context subset per content type — LinkedIn gets bio + philosophy; Bio gets case studies.

prompt_templates.py

5 template functions, one shared voice system prompt. Uniqueness test built into every template: "Could generic AI write this?"

content_pipeline.py

Orchestrates all modules. Routes by tone and content type. Auto-saves .md outputs with timestamp, appends to cost_log.md.

Data & Research

llm_integration.py

OpenAI GPT-4o-mini wrapper. Token counting, cost tracking per request. Session summary. ~\$0.0003 per LinkedIn post.

web_researcher.py

Searches HN Algolia, Dev.to, TechCrunch, InfoQ RSS. max_results passed to every source — no silent hardcoding. Under 10s.

hn_processor.py

HackerNews Algolia API. Topic-filtered keyword search or top AI stories by score. Formats stories for Commentary tab.

Interface & Management

kb_manager.py

KB read/write layer. 6 categories auto-route to the right file. New entries available to the next generation immediately.

gradio_app.py

6-tab web UI: LinkedIn Post, Executive Bio, AI Commentary (3 modes, 2 sources), Leadership, KB Management, Cost & Quality.

trello_manager.py

Trello REST API. Automated board setup, FR card creation with WIP limit enforcement. Project management integration.

Demo Evidence — Same topic, three genuinely different outputs



Personal Story

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. Not generic AI — your voice, your frameworks, your proof points.

LinkedIn Post Executive Bio AI Commentary Leadership Signature Cost & Quality

Three genuinely different modes:

- Personal Story — draws from your KB (via case studies, frameworks)
- Industry Observation — searches the web, synthesizes what others are saying
- Thought Leadership — industry context + your personal experience anchored in KB

Topic or angle

why engineering orgs fail when the culture is not well established

Tone

personal story industry observation thought leadership

Generate LinkedIn Post

Generated post

A startling observation over 70% of engineering organizations fail to reach their potential, not because of technical focus, but due to weak culture.

In my 20+ years leading transformations at companies like Delivery Hero and semder, I've seen firsthand how an underdeveloped culture can derail even the most technically proficient teams. At Delivery Hero, we faced a staggering 20% incidents per week. It wasn't just technical challenges; it was a cultural one. By implementing the Quality (Maturity Index QMI), we not only reduced engineering health but also made the invisible problems visible. Within six months, 60% of the organization voluntarily adopted it, driving accountability and improvement.

At semder, I authored the Engineering (Meta) to establish our values and practices. This living document ensured that every engineer had clarity about our collective goals before they wrote their first line of code. The result? A more cohesive team that understood how to prioritize both product features and technical debt, leading to a reduction in incidents to just 2 per month.

Engineering culture isn't just a feel-good concept; it's the infrastructure that supports our technical goals. When culture thrives, so do the outcomes—like the 35% organic growth we achieved through AI recommendations.

Have you experienced a culture transformation in your engineering organization? What did you learn?

Saved to

Users\kenneth\Downloads\Project_AIContentCreator\output\linkedin_post_20260528_194733.md

Cost

Cost: \$0.00001 | Session: \$0.0028 | Tokens: 15,224

Quality metrics

CONTENT QUALITY METRICS

Type:	linkedin_post
Tone:	personal_story
Words:	227
Sentences:	15
Avg sentence len:	14.7 words
Vocabulary richness:	68.8%
Readability:	38.4 (Complex)
Specificity score:	20 (numbers/proper nouns)
Personal refs:	3 (KB grounding)
Web sources used:	0
Generation time:	10.76s

Uniqueness vs generic AI: 8.4% similarity

Very unique (Saved to quality_log.md)



Industry Observation

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LinkedIn Post Executive Bio AI Commentary Leadership Signature Cost & Quality

Three genuinely different modes:

- Personal Story — draws from your KB (via case studies, frameworks)
- Industry Observation — searches the web, synthesizes what others are saying
- Thought Leadership — industry context + your personal experience anchored in KB

Topic or angle

why engineering orgs fail when the culture is not well established

Tone

personal story industry observation thought leadership

Fetch industry context (5-10s)

Industry sources found

1. [TechCunch]
Startup Boltified 200 application deadline extended to June 8 after overwhelming demand
After several high-demand firms founded around the world, TechCunch has extended the Startup Boltified 200 application deadline to June 8. Finally.

2. [TechCunch]
Corp announces \$100M raise at \$2.6B valuation — double what it was worth three weeks ago
Insurance tech Corp announced today an \$100-million Series B raise, valuing the company at \$2.6 billion, just three weeks after announcing a \$60M raise.

Saved to

Users\kenneth\Downloads\Project_AIContentCreator\output\linkedin_post_20260528_192755.md

Cost

Cost: \$0.00028 | Session: \$0.003 | Tokens: 138

Quality metrics

CONTENT QUALITY METRICS

Type:	linkedin_post
Tone:	industry_observation
Words:	25
Sentences:	1
Avg sentence len:	19.5 words
Vocabulary richness:	63.8%
Readability:	30.8 (Complex)
Specificity score:	5 (numbers/proper nouns)
Personal refs:	0 (KB grounding)
Web sources used:	2
Generation time:	7.04s

Uniqueness vs generic AI: 8.4% similarity

Very unique (Saved to quality_log.md)



Thought Leadership

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LinkedIn Post Executive Bio AI Commentary Leadership Signature Cost & Quality

Three genuinely different modes:

- Personal Story — draws from your KB (via case studies, frameworks)
- Industry Observation — searches the web, synthesizes what others are saying
- Thought Leadership — industry context + your personal experience anchored in KB

Topic or angle

why engineering orgs fail when the culture is not well established

Tone

personal story industry observation thought leadership

Fetch industry context (5-10s)

Industry sources found

1. [TechCunch]
Why Paris may be the most important AI city outside Silicon Valley
Europe's startup ecosystem has matured significantly; its founders are increasingly willing to scale companies domestically instead of immediately exiting.

2. [TechCunch]
In just 3 weeks, Stricq AI is coming to Los Angeles
Stricq AI, Los Angeles is on June 18, join for meaningful networking and freddie chats with leaders from Mach Industries, Shrike Systems, and more...

Saved to

Users\kenneth\Downloads\Project_AIContentCreator\output\linkedin_post_20260528_194203.md

Cost

Cost: \$0.00079 | Session: \$0.006 | Tokens: 8263

Quality metrics

CONTENT QUALITY METRICS

Type:	linkedin_post
Tone:	thought_leadership
Words:	209
Sentences:	13
Avg sentence len:	16.1 words
Vocabulary richness:	72.7%
Readability:	24.5 (Complex)
Specificity score:	13 (numbers/proper nouns)
Personal refs:	7 (KB grounding)
Web sources used:	2
Generation time:	8.02s

Uniqueness vs generic AI: 8.5% similarity

Very unique (Saved to quality_log.md)

Topic: “Why engineering orgs fail when culture is not well established” — Personal refs: 9 (KB) / 7 (KB) / 0 (web) — Uniqueness vs generic AI: ~8.5% similarity ✓ Very unique all three

Mode Comparison — This engine vs ChatGPT, same prompt

Dimension	 Personal Story	 Industry Obs.	 Thought Leader.	ChatGPT (Generic)
Source	KB only	Web-first	KB + Web	Generic LLM
Personal refs	9 (KB grounded)	0 (web only)	7 (KB grounded)	0 ✗
Web sources	0	2 (live news)	2 (live news)	0 ✗
Voice / angle	1st person narrative	Analyst / trend observer	Opinionated hybrid	Instructional listicle
Uniqueness	8.4% similar ✓	8.4% similar ✓	8.5% similar ✓	Baseline ✗
LinkedIn-ready?	✓ Yes, as-is	✓ Yes, timely hook	✓ Yes, strong POV	✗ Needs rewrite

The key story: ChatGPT produced an 800-word instructional essay — no voice, no specifics, zero proof points, not LinkedIn-appropriate. This engine produced 3 genuinely different, platform-ready posts from the same prompt — each with a distinct angle, real grounding, and under 10 seconds.

Generation time	10.13s	7.04s	8.09s	5.04s
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Executive Bio — 6 purpose-adapted variants, one knowledge base

Short / General

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. The generic AI — your voice, your framework, your proof points.

Unleash Pro AI Executive Pro AI Commentary Leadership Signature Cost & Quality

Length: short medium long

Purpose: general **speaking** board job application

Generate Bio

Saved to: /Users/annemaria/Downloads/Project2_AICreator/contents/output/Executive_Bio_20250208_1541.md

Cost: Cost: \$0.000862 | Session: \$0.0001 | Tokens: 25456

Quality metrics

COPIRETT QUALITY METRICS

Type: executive bio
Tone: professional
Uniqueness: 92%
Readability: 92%
Specificity: 92% (9 numbers/keyword round)
Personal info: 9/10 (grounding)
Link sources used: 0
Generation time: 3.5s

Uniqueness in generic: 92.42% similarity
Very unique
(Clicked to quality, log md)

Short / Speaking

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. The generic AI — your voice, your framework, your proof points.

Unleash Pro AI Executive Pro AI Commentary Leadership Signature Cost & Quality

Length: short medium long

Purpose: general **speaking** board job application

Generate Bio

Saved to: /Users/annemaria/Downloads/Project2_AICreator/contents/output/Executive_Bio_20250208_1544.md

Cost: Cost: \$0.000862 | Session: \$0.0001 | Tokens: 31554

Quality metrics

COPIRETT QUALITY METRICS

Type: executive bio
Tone: professional
Uniqueness: 92%
Readability: 92%
Specificity: 92% (9 numbers/keyword round)
Personal info: 9/10 (grounding)
Link sources used: 0
Generation time: 3.5s

Uniqueness in generic: 92.42% similarity
Very unique
(Clicked to quality, log md)

Short / Board

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. The generic AI — your voice, your framework, your proof points.

Unleash Pro AI Executive Pro AI Commentary Leadership Signature Cost & Quality

Length: short medium long

Purpose: general **speaking** **board** job application

Generate Bio

Saved to: /Users/annemaria/Downloads/Project2_AICreator/contents/output/Executive_Bio_20250208_154245.md

Cost: Cost: \$0.000862 | Session: \$0.0001 | Tokens: 18433

Quality metrics

COPIRETT QUALITY METRICS

Type: executive bio
Tone: professional
Uniqueness: 92%
Readability: 92%
Specificity: 92% (9 numbers/keyword round)
Personal info: 9/10 (grounding)
Link sources used: 0
Generation time: 4.37s

Uniqueness in generic: 92.42% similarity
Very unique
(Clicked to quality, log md)

Short / Job Application

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. The generic AI — your voice, your framework, your proof points.

Unleash Pro AI Executive Pro AI Commentary Leadership Signature Cost & Quality

Length: short medium long

Purpose: general **speaking** board **job application**

Generate Bio

Saved to: /Users/annemaria/Downloads/Project2_AICreator/contents/output/Executive_Bio_20250208_160201.md

Cost: Cost: \$0.000862 | Session: \$0.0001 | Tokens: 24434

Quality metrics

COPIRETT QUALITY METRICS

Type: executive bio
Tone: professional
Uniqueness: 92%
Readability: 92%
Specificity: 92% (9 numbers/keyword round)
Personal info: 9/10 (grounding)
Link sources used: 0
Generation time: 3.5s

Uniqueness in generic: 92.42% similarity
Very unique
(Clicked to quality, log md)

Long / General

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. The generic AI — your voice, your framework, your proof points.

Unleash Pro AI Executive Pro AI Commentary Leadership Signature Cost & Quality

Length: short medium long

Purpose: general **speaking** board job application

Generate Bio

Saved to: /Users/annemaria/Downloads/Project2_AICreator/contents/output/Executive_Bio_20250208_155442.md

Cost: Cost: \$0.000862 | Session: \$0.0001 | Tokens: 42438

Quality metrics

COPIRETT QUALITY METRICS

Type: executive bio
Tone: professional
Uniqueness: 92%
Readability: 92%
Specificity: 92% (9 numbers/keyword round)
Personal info: 9/10 (grounding)
Link sources used: 0
Generation time: 3.5s

Uniqueness in generic: 92.42% similarity
Very unique
(Clicked to quality, log md)

Long / Board

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. The generic AI — your voice, your framework, your proof points.

Unleash Pro AI Executive Pro AI Commentary Leadership Signature Cost & Quality

Length: short medium long

Purpose: general **speaking** **board** job application

Generate Bio

Saved to: /Users/annemaria/Downloads/Project2_AICreator/contents/output/Executive_Bio_20250208_155442.md

Cost: Cost: \$0.000862 | Session: \$0.0001 | Tokens: 36770

Quality metrics

COPIRETT QUALITY METRICS

Type: executive bio
Tone: professional
Uniqueness: 92%
Readability: 92%
Specificity: 92% (9 numbers/keyword round)
Personal info: 9/10 (grounding)
Link sources used: 0
Generation time: 3.5s

Uniqueness in generic: 92.42% similarity
Very unique
(Clicked to quality, log md)

All 6 variants grounded in the same KB — Personal refs: 8-18 per bio — Uniqueness vs generic AI: 6.7-13.3% similarity ✓ Very unique — Cost: ~\$0.0009 per bio

Bio Comparison — This engine vs ChatGPT (with full context), same person

Dimension	 Engine (Short)	 Engine (Long)	ChatGPT
Purpose-adapted?	✅ 4 variants	✅ 2 variants	❌ One size
Word count	99-107 words	348-383 words	~350 words
Personal refs (KB)	8-10 per variant	18 per variant	0 ❌
Proof points	QMI, 20→2, 35% growth	All KPIs + AUDI A8	Generic patterns ❌
Uniqueness	6.7-11% similar ✅	12.6-13.3% similar ✅	Baseline ❌
Generation time	2.54s - 4.37s	9.81s - 10.85s	3.07s

The structural difference: ChatGPT writes about Ioanna in abstract leadership language — no numbers, no real decisions, no proof points. It reads like a LinkedIn summary written from a job title.
This engine writes from evidence — and adapts the same story for 6 different audiences, automatically.



AI Commentary — Week 2 redesign: 3 modes, 2 news sources

❌ Original behaviour

- Single mode only — always anchored to KB
- MIT AI lecture videos (2016) → generated QMI reference
- HackerNews only — top AI stories, no topic filter
- Web sources listed in footer but not accessible in UI
- max_results parameter silently ignored — hardcoded at 3/3/2/2

✅ Redesigned

 **Story Summary**
Faithful, no KB forced

 **Personal Commentary**
KB only if genuinely relevant

 **Multi-Story Synthesis**
Cross-story trend narrative

 **Browse HackerNews**

- Top AI stories by score
- Optional topic filter (new)
- Multi-select dropdown

 **Search by Topic**

- HN + Dev.to + TechCrunch + Web
- CheckboxGroup — Select All / Clear
- max_results bug fixed in WebResearcher

AI Content Creator — Personal Brand Engine

Generates unique, evidence-based content. Not generic AI — your voice, your frameworks, your proof points.

[LinkedIn Post](#) [Executive Bio](#) [AI Commentary](#) [Leadership Signature](#) [Knowledge Base](#) [Cost & Quality](#)

Generation mode — choose what kind of output you want:

- **Story Summary** — faithful overview, no personal references forced
- **Personal Commentary** — overview first, then Joanna's angle *only if there is a genuine connection*
- **Multi-Story Synthesis** — select 2-5 stories, one trend narrative

Generation mode

☒ Story Summary ☐ Personal Commentary

☒ Multi-Story Synthesis

News source

☐ Browse HackerNews

☒ Search by Topic

Search by topic

Topic to search

AI-assisted development opinion

Sources

☐ HackerNews

☐ Dev.to

☐ TechCrunch

☒ Generate Web

Max results (actual count depends on source availability)

1 5 10

Textbox

✓ 1 results — 1 from web

Select results to include (multi-select supported)

☒ [Packer News] Show #1: [MatSeeker - a map of what is safe to change in large codebases]

Select All

Clear

News item(s)

AI Commentary

Generated commentary

MatSeeker introduces a tool designed to help developers navigate large codebases by identifying which areas are safe to modify and which are not. This initiative addresses a common pain point in software development, while fixing bugs is often straightforward, knowing where to make changes without introducing new issues can be daunting.

In a landscape where codebases are continually evolving and project complexity is increasing, tools like MatSeeker are becoming essential. They not only enhance developer efficiency but also mitigate the risk of unintended side effects from code changes. As engineering teams face mounting pressures to deliver rapid updates while maintaining system integrity, solutions that provide insights into code safety are invaluable.

As we embrace more sophisticated technologies, the challenge of managing large codebases will only grow. How can your engineering team leverage tools like MatSeeker to improve code quality and agility in development?

Saved to

/Users/joannamenta/Downloads/Project2_AIContentCreator/output/ai_commentary_20260530_123527.md

Cost

Cost: \$0.00031 | Session: \$0.0006 | Tokens: 2,223

Quality metrics

CONTENT QUALITY METRICS

Type: ai_commentary
Tone: commentary/summary
Words: 343
Sentences: 8
Avg sentence len: 17.9 words
Vocabulary richness: 74.8%
Readability: 23.0 (Complex)
Specificity score: 0 (numbers/proper nouns)
Personal refs: 0 (KB grounding)
Web sources used: 0
Generation time: 2.38s

Uniqueness vs generic AI: 6.7% similarity
+ 100% unique
(Saved to quality_log.md)

Design principle: Faithfulness to the source matters more than forced personal anchoring. The KB is a resource, not a mandatory ingredient in every output.

Knowledge Base management (kb_manager.py) was also shipped in Week 2 — covered in the Completion Status slide.

Knowledge Base Management — update at any stage — no restart needed

AI Content Creator — Personal Brand Engine

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📄 LinkedIn Post 📄 Executive Bio 📄 AI Commentary 📄 Leadership Signature 📄 Knowledge Base 📄 Cost & Quality

Add new entries or browse existing knowledge base files.

All changes are immediately available to the next generation — no restart needed. The pipeline reads KB files fresh on every call.

+ Add New Entry

Use this to add a new story, achievement, observation, or insight directly — without going through a content tab.

Category

Case Study

Target file (auto-selected from category)

primary / case_studies.md

Content

When Product Management struggled to create prototypes because Engineering did not give them Priority, instead of just accepting the situation, Ioanna has introduced librechat as an OSS interface and formulated a small task force to find a way to create an easy way for PTHs to query DBs and start with that. the team established MCP servers and agents fed with the figma files required to match the look and feel of the company and the first prototype was able to be achieved without engineering involvement within one week.

Save Entry

Save status

✓ Saved to knowledge_base/primary/case_studies.md (524 chars) — available in next generation.

Browse Knowledge Base

Read-only view. Edit files directly in your editor for structural changes.

KB

File

Refresh

primary

secondary

File content

```
## Case Study 6: AI Tooling Adoption at sennder
**Company:** sennder
**Role:** VP Engineering, Platform & Data
**Situation:** The organisation needed to adopt AI tooling at scale without creating security risks, governance gaps, or a two-tier system where some teams had AI and others didn't.
**Actions:**
- Deployed LibreChat as the internal AI interface — giving engineers access to LLM capabilities within a governed environment
- Built MCP server integration for commercial email parsing (OCR pipeline)
- Established AI usage guidelines as part of the Engineering Manifesto
- Rolled out AI tooling consistently across all teams
**Outcome:** Organisation-wide AI tooling adoption with governance in place from day one.
**Leadership lesson:** AI adoption without governance creates technical debt at AI speed. Define how you use AI before you use it everywhere.

## Quantified Impact Summary
| Metric | Before | After | Context |
| --- | --- | --- | --- |
| P1 incidents | 20/week | 2/month | sennder |
| Navigation cold-start | 55 seconds | 5 seconds | HERE/AUDI A8 |
| QM1 adoption | 0% | 50% of org | Delivery Hero |
| Organic order growth | baseline | +35% | Delivery Hero (ML) |
| Customer retention | baseline | +60% | Delivery Hero (ML) |
| Monolith migration | in progress | complete | sennder (6 months) |
```

Powered by OpenAI GPT-4o-mini · KB: primary + secondary · Live news: HackerNews + Dev.to + TechCrunch

+ Add New Entry

- 6 categories auto-route to the right KB file
- Target file shown before you save
- Save confirmation with char count

📄 Browse KB

- Read any KB file inline
- Switch primary / secondary at any time
- Refresh to see latest content

📄 Save to KB (any tab)

- Every output tab has a Save to KB button
- Pre-fills content from generated output
- Editable before confirming

The pipeline reads KB files fresh on every call — new entries are available to the next generation with zero restart.

Challenges & Learnings



Making content genuinely unique

Built knowledge base from real data, not placeholders. Added uniqueness test to every prompt template: "Could generic AI write this without knowing Ioanna's background?"



HackerNews API — 2-minute waits

Original Firebase item-by-item API scanned 100 stories = ~140 seconds. Switched to Algolia search API — one call returns 5 relevant stories in under 2 seconds.



Context selection per content type

Not all documents are relevant for all content types. Built `get_context_for_content_type()` to map each type to the most relevant document subset.



Cost visibility

Cost tracker built in from day one. Every generation appends to `cost_log.md`. Session total shown inline after every output. ~\$0.002 for a full 4-type session.

Design Decisions — What looked simple, wasn't

Decision	Options considered	What we chose → why
Where does “Save to KB” live?	• Accordion bottom of each tab • Dedicated KB tab only • One-click button per tab + full tab	✅ Hybrid (option 3): save where you generate, manage in one place. Friction-free capture matters.
Commentary: how many modes and what do they produce?	• One mode (always personal) • Personal toggle checkbox • Three distinct radio modes	✅ Three modes: summary / personal / synthesis are genuinely different outputs, not variations of tone.
Multi-select for search results	• gr Dropdown(multiselect=True) — expand arrow broken in Gradio 6.x • Single select + editable textbox • CheckboxGroup + Select All	✅ CheckboxGroup: reliable across Gradio versions, Select All button natural to add, no arrow dependency.
Results slider set to 7 — returns 3. Why?	• Web researcher has internal cap • max_results accepted but not passed to sources — hardcoded 3/3/2/2	✅ Fixed in web_researcher.py: max_results passed to every source. Deduplication caps the final list.
When is output “good enough” to stop iterating?	• Sounds unique but has no personal proof points • Has proof points but reads like a CV • Platform-ready + specific + authentic	✅ Uniqueness test built into every prompt: “Could generic AI write this without knowing Ioanna’s background?”

Every design decision above was real — discovered by running the system, not by theorising. The UI is a product of iteration, not a plan.

Week 2 Completion Status



Completed in Week 2

- ✓ HackerNews API integration (live feed)
- ✓ Web-grounded LinkedIn posts (Industry Observation + Thought Leadership)
- ✓ AI Commentary redesign — 3 modes, no forced KB anchoring
- ✓ Multi-source search (HN + Dev.to + TechCrunch + General Web)
- ✓ Knowledge Base management — Save to KB + Browse tab
- ✓ Uniqueness comparison vs ChatGPT (LinkedIn + Bio)
- ✓ Prompt refinement + max_results bug fix in web_researcher.py
- ✓ README polish + GitHub push



Future vision

- Notion integration — content calendar workflow
- Multi-user white-label (swap KB folder per user)
- RAG layer for larger knowledge bases
- LinkedIn API — optional automated publishing
- AI Commentary screenshots + live demo

The product insight

The white-label version is just a folder swap. The hard work — prompt engineering, pipeline architecture, knowledge base structure — is already done.

Cost at scale: ~\$0.002 per full session • GPT-4o-mini \$0.15/M input tokens

Thank You

Questions?

GitHub: github.com/ioannarenta

Trello: trello.com/b/kteKFdzb

Course: Ironhack

Ioanna Renta · VP Engineering · Berlin